Week 7 Lab 2 – Identifying IPv4 Addresses

Objectives

Part 1: Identify IPv4 Addresses

- Identify the network and host portion of an IP address.
- Identify the range of host addresses given a network/prefix mask pair.

Part 2: Classify IPv4 Addresses

- Identify the type of address (network, host, multicast, or broadcast).
- Identify whether an address is public or private.
- Determine if an address assignment is a valid host address.

Background / Scenario

Addressing is an important function of network layer protocols because it enables data communication between hosts on the same network, or on different networks. In this lab, you will examine the structure of Internet Protocol version 4 (IPv4) addresses. You will identify the various types of IPv4 addresses and the components that help comprise the address, such as network portion, host portion, and subnet mask. Types of addresses covered include public, private, unicast, and multicast.

Subnet Masks are usually stated in a “dotted decimal” format similar to IP addresses. However, because the mask is always ones to the left and zeros to the right, it can be convenient to convey the mask using a “slash number”, the number identifies the number of bits (binary ones) in the mask.

Some systems will allow addresses to be entered with a slash number, some will not and some will insist on it. You need to be able to convert between a subnet mask and a slash number in either direction.

This tutorial is not easy, take your time and repeat earlier work / tutorials if necessary.

Required Resources

- Device with Internet access
- Optional: IPv4 address calculator (same warning as before, you will only have a basic calculator in the exam)

Part 1: Identify IPv4 Addresses

In Part 1, you will be given several examples of IPv4 addresses and will complete tables with appropriate information.

Step 1: Analyze the table shown below and identify the network portion and host portion of the given IPv4 addresses.

The first two rows show examples of how the table should be completed.

Key for table:

N = all 8 bits of an octet are in the network portion of the address

n = a bit in the network portion of the address

H = all 8 bits of an octet are in the host portion of the address

h = a bit in the host portion of the address

IP Address/Prefix	Network/Host N,n = Network H,h = Host	Subnet Mask	Network Address
192.168.10.10/24	N.N.N.H	255.255.255.0	192.168.10.0
10.101.99.17/23	N.N.nnnnnnnh.H	255.255.254.0	10.101.98.0
209.165.200.227/27			
172.31.45.252/24			
10.1.8.200/26			
172.16.117.77/20			
10.1.1.101/25			
209.165.202.140/27			
192.168.28.45/28			

Remember the rules given in the lecture e.g. network addresses must always begin as an even number (in the significant octet), broadcast addresses are always odd numbers!

Step 2: Analyze the table below and list the range of host and broadcast addresses given a network/prefix mask pair.

The first row shows an example of how the table should be completed.

IP Address/Prefix	First Host Address	Last Host Address	Broadcast Address
192.168.10.10/24	192.168.10.1	192.168.10.254	192.168.10.255
10.101.99.17/23			
209.165.200.227/27			
172.31.45.252/24			
10.1.8.200/26			
172.16.117.77/20			
10.1.1.101/25			
209.165.202.140/27			
192.168.28.45/28			

Part 2: Classify IPv4 Addresses

In Part 2, you will identify and classify several examples of IPv4 addresses. You may need to research the identity of Public, Private and Multicast IPv4 addresses

A broadcast address is the “all ones” address in a subnet, the network address is the “all zeros” address in the subnet. Neither of these special addresses can be used as host addresses.

Step 1: Analyse the table shown below and identify the type of address (network, host, multicast, or broadcast address).

The first row shows an example of how the table should be completed.

IP Address	Subnet Mask	Address Type
10.1.1.1	255.255.255.252	host
192.168.33.63	255.255.255.192	
239.192.1.100	255.252.0.0	
172.25.12.52	255.255.255.0	
10.255.0.0	255.0.0.0	
172.16.128.48	255.255.255.240	
209.165.202.159	255.255.255.224	
172.16.0.255	255.255.0.0	
224.10.1.11	255.255.255.0	

Step 2: Analyse the table shown below and identify the address as public or private.

IP Address/Prefix	Public or Private
209.165.201.30/27	
192.168.255.253/24	
10.100.11.103/16	
172.30.1.100/28	
192.31.7.11/24	
172.20.18.150/22	
128.107.10.1/16	
192.135.250.10/24	
64.104.0.11/16	

Step 3: Analyse the table shown below and identify whether the address/prefix pair is a valid host address.

IP Address/Prefix	Valid Host Address?	Reason
127.1.0.10/24		
172.16.255.0/16		
241.19.10.100/24		
192.168.0.254/24		
192.31.7.255/24		
64.102.255.255/14		
224.0.0.5/16		
10.0.255.255/8		
198.133.219.8/24		

Reflection

Why should we continue to study and learn about IPv4 addressing if the available IPv4 address space is depleted?

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Congratulations for reading this far! Some shortcuts that might be helpful to “rule things out”:

Using “all ones” means that the “last bit” of the last octet must be set. The “one” bit alone determines whether a number is odd or even. A broadcast therefore is always an odd number, this does *not* mean that all odd number addresses are broadcast addresses but it *does* mean that even number addresses *cannot* be broadcast addresses.

Similarly using “all zeros” means that the “last bit” of the last octet cannot be set and, as a result, the number must be even – an odd number cannot be a network address.

A subnet mask is ones to the left, zeros to the right. The only number that can have the one bit set in any octet is the “all ones” for that octet. The only odd number that can be part of a subnet mask is 255.

The only time that any subnet mask octet can contain any number other than 0 is if the preceding octet is 255 (all ones), any other number means that a zero has been used and this means that all subsequent bits must be zero.

The network address and broadcast address cannot be used as host addresses.

The first host address is the network address + 1. The network address is always even so the first host must be odd.

The last network address is the broadcast address – 1. The broadcast address is always odd so the last host address must be even.

Don’t forget that the address zero exists! 3 bits is the range 000b to 111b (0d to 7d), that is eight addresses not seven.